

IF YOU CAN'T TRUST, THEN VERIFY

How the Glass Perimeter could enable AI treaties

THE OVERHANG · WASHINGTON DC · SEPTEMBER 19, 2026



CANARY
INSTITUTE
FOR AI POLICY

THE FRAME HAS MOVED



TWO WEEKS THAT SHIFTED THE CONVERSATION

- Coxon triggered a preference cascade on existential risk
- UN: “the last generation able to set the terms”
- Anthropic pledged unilateral external audits — OpenAI agreed to follow
- **A deal you can’t verify is a deal you can’t keep.**



AMERICAN INSPECTORS, SOVIET GATES

TWO DECADES OF COOPERATION, ZERO TRUST

- American inspectors stood at Soviet missile factories for twenty years
- Every container big enough to hide a banned missile got x-rayed
- Russian inspectors did the same in Utah
- Nobody trusted anybody — the treaty held anyway
- **Verification is what makes cooperation possible in the absence of trust.**



Soviet inspectors, Pershing II elimination, Jan 1989 · U.S. DoD photo (public domain)



ONE CHOKEPOINT, ONE FIELD

THE TRACKING MACHINERY IS HALF-BUILT

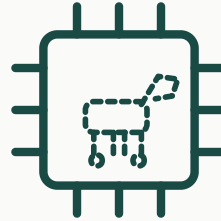
- Fissile material ↔ leading-edge chips: one scarce physical resource, one field
- EUV lithography = ASML, one company on Earth
- Leading-edge fabrication = TSMC and a handful of others
- Dual-use technology → aim at verification, not seizure
- **The chokepoints are already there. Export controls half-built the machinery.**



ASML machine containers, Tainan, May 2025 · 4300streetcar / Wikimedia Commons, CC BY 4.0



SILICON WE BOTH TRUST DOESN'T EXIST



FORTY YEARS OF ARMS TALKS, NO SHARED CHIP

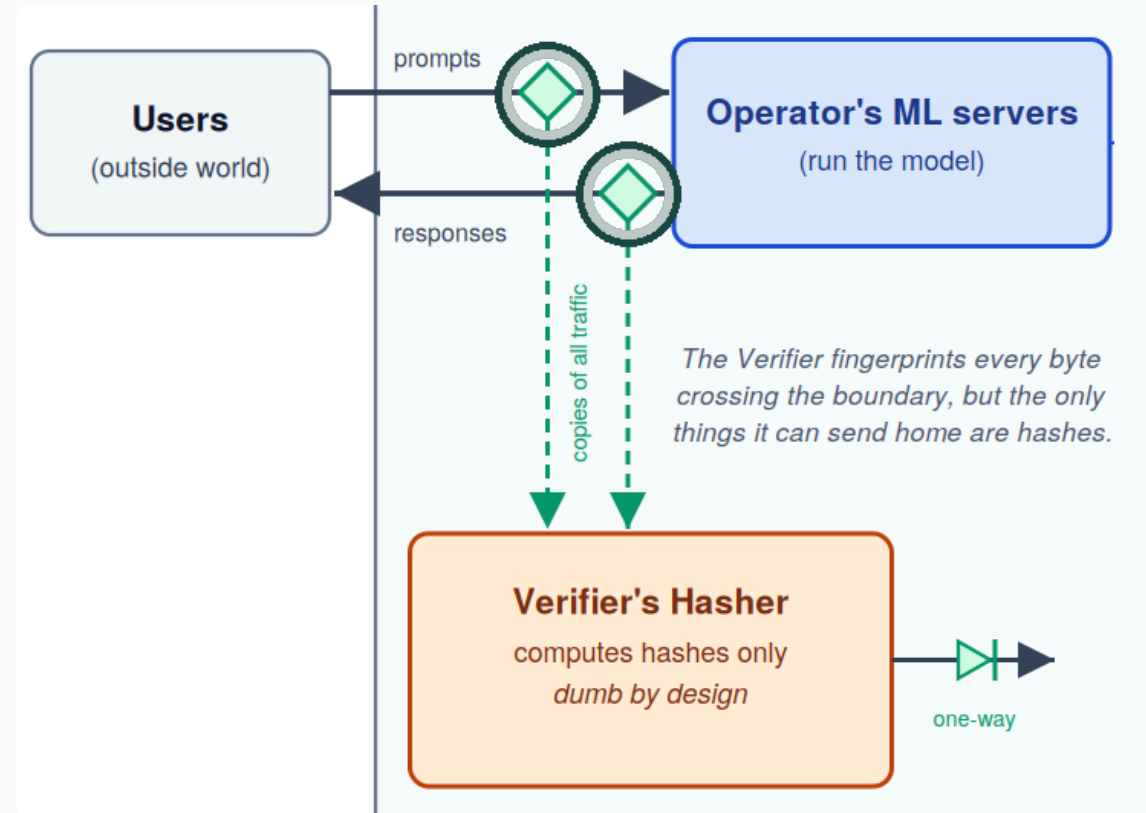
- Hardware trojans in dopants: invisible to optical inspection
- Rowhammer: reading memory rows in rhythm flips bits in others — pure physics
- No accepted method for proving a chip free of hidden functionality
- Each side trusts its own silicon; neither trusts the other's
- **Any verification regime has to work without mutually-trusted chips.**



PHOTONS CAN'T BE PROGRAMMED

THE GLASS PERIMETER · AFTER CANKAYA, 2026

- Optical splitter: passive glass, no electronics, physically can't lie
- Every prompt in, every response out — both parties see the same photons
- Each side processes on its own hardware, on its own soil
- **The boundary between adversaries is literally glass. Their electronics never touch yours.**



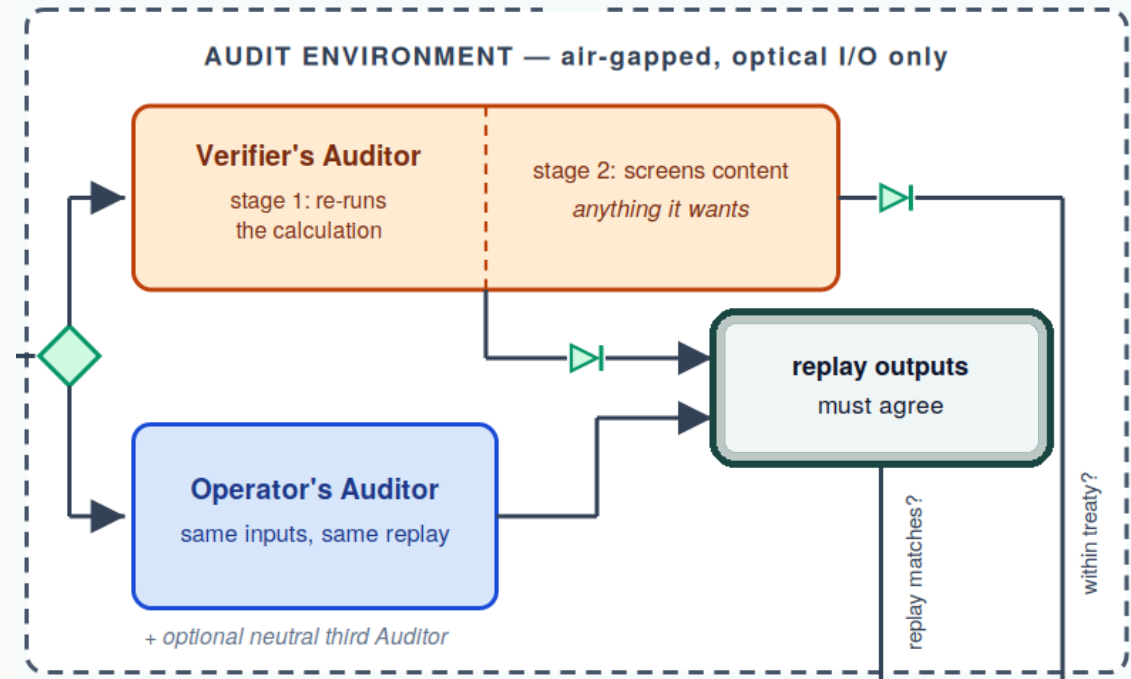
From "If you can't trust, then verify" (canaryinstitute.ai) · splitters highlighted



ENTER THE SEALED ROOM

RANDOM SPOT-CHECKS, MATCHED REPLAYS

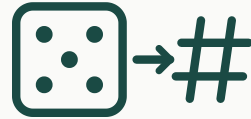
- Verifier picks an output at random — a coin flip is fine
- Weights, input, and manifest enter an air-gapped Audit Environment
- Both sides replay the calculation; both compute the hash
- Auditor emits one signed message: within-treaty, hashes match, done
- **Model weights go in. Only a signed verdict comes out.**



From "If you can't trust, then verify" (canaryinstitute.ai) · Audit Environment detail



RANDOM BEATS EXHAUSTIVE

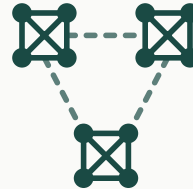


YOU DO NOT HAVE TO AUDIT EVERYTHING

- Zero failures in n random audits $\rightarrow$ 95% confidence $< 3/n$ of traffic is dirty
- 300 audits/day $\rightarrow$ cheating bounded below 1% of workload
- 10,000 audits/day $\rightarrow$ cheating bounded below 0.03%
- The bound doesn't depend on datacenter size
- **Cheating becomes almost impossible to hide.**



INFERENCE IS EASY. TRAINING IS THE FRONTIER.



WHERE THE WORK STILL IS

- Training: every chip talks to every chip — no clean I/O boundary
- DiLoCo / Streaming DiLoCo: islands sync occasionally, deterministically
- Bit-exact replay across GPU generations: ~85% today, needs 100%
- **The engineering is hard. None of it requires a fundamental breakthrough.**



TWENTY-FIVE PEOPLE



THE LOAD-BEARING FIELD, AND HOW FEW ARE IN IT

- ~60 papers total in the entire field
- ~72 researchers who've published; most are part-time
- ~25 full-time equivalents worldwide
- All accumulated knowledge = one week of reading
- **Twenty-five people stand between us and enforceable AI treaties.**



SIX PROBLEM AREAS, ALL OPEN

HARDWARE & OPTICS

800G passive taps, tamper-evident enclosures, fast memory wiping

NETWORKING & SYSTEMS

Line-rate packet hashing, detecting distributed training on the wire

CRYPTOGRAPHY

ZK proofs of inference/training at frontier scale, bit-exact replay

RED-TEAMING

Nobody is red-teaming the whole regime yet — TEE, HEM, spoofing

PHYSICAL EVIDENCE

Power traces, thermal signatures, satellite imagery, grid accounting

POLICY & INSTITUTIONS

Attribution ladders, deal-stability, the international body itself

LIVING PROJECT LISTS: [PROOFWORKS](#) · [AI-2040](#) · [AMODO](#) · [COMPUTE VERIFICATION PROJECT](#)

Six months of your work = a couple percentage points of the field. That's HUGE.



IF WE WANT TO, WE CAN

WHETHER WE PACE THE FRONTIER ISN'T A
TECHNOLOGY QUESTION.
IT'S A QUESTION OF WHETHER WE WANT TO.

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